Name:M.Veeramathivathani

Register no:192512255

Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Aim:

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

Objective – 3: Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance

Software Required:

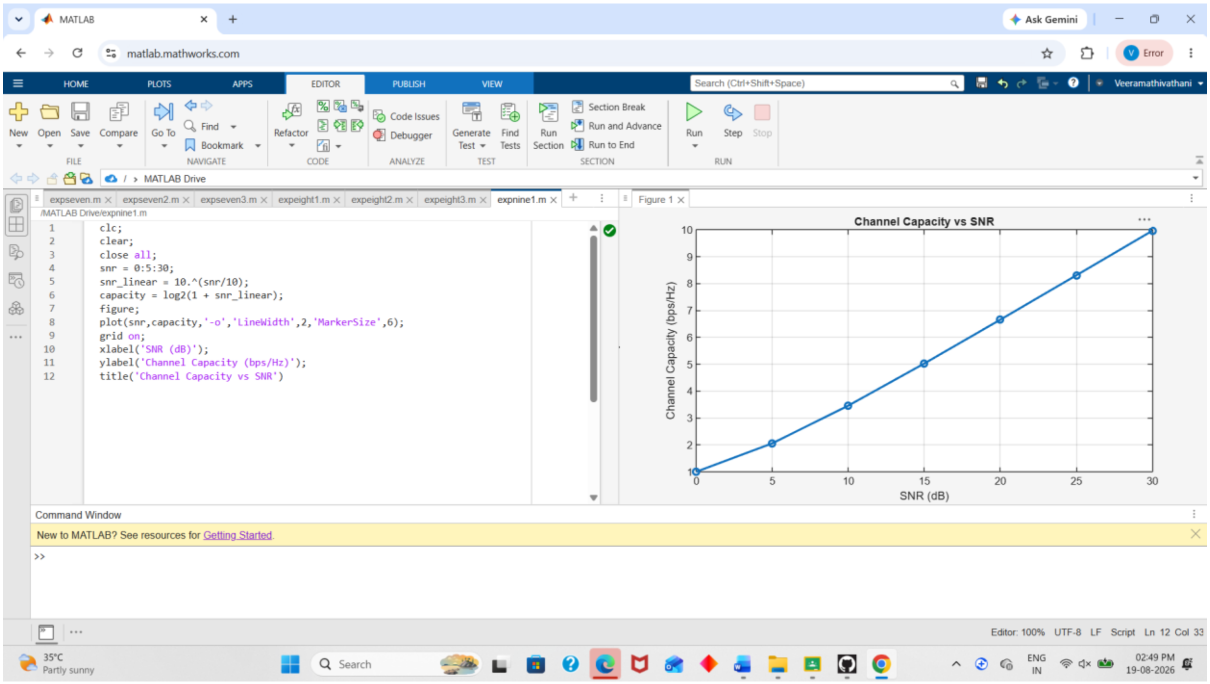
• MATLAB

Theory

Massive MIMO employs a large number of antennas at the base station to simultaneously serve multiple users. The increased spatial diversity and beamforming capability significantly improve channel capacity, signal quality, and antenna gain. Channel capacity is influenced by the signal□to-noise ratio (SNR), while antenna gain increases with the number of antenna elements. These parameters are fundamental to achieving the high throughput and spectral efficiency required in 5G and beyond wireless communication systems.

Objective 1: Matlab Code and Output

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR')
```



Objective 2: Matlab Code and Output

```
clc;

clear;

close all;
```

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

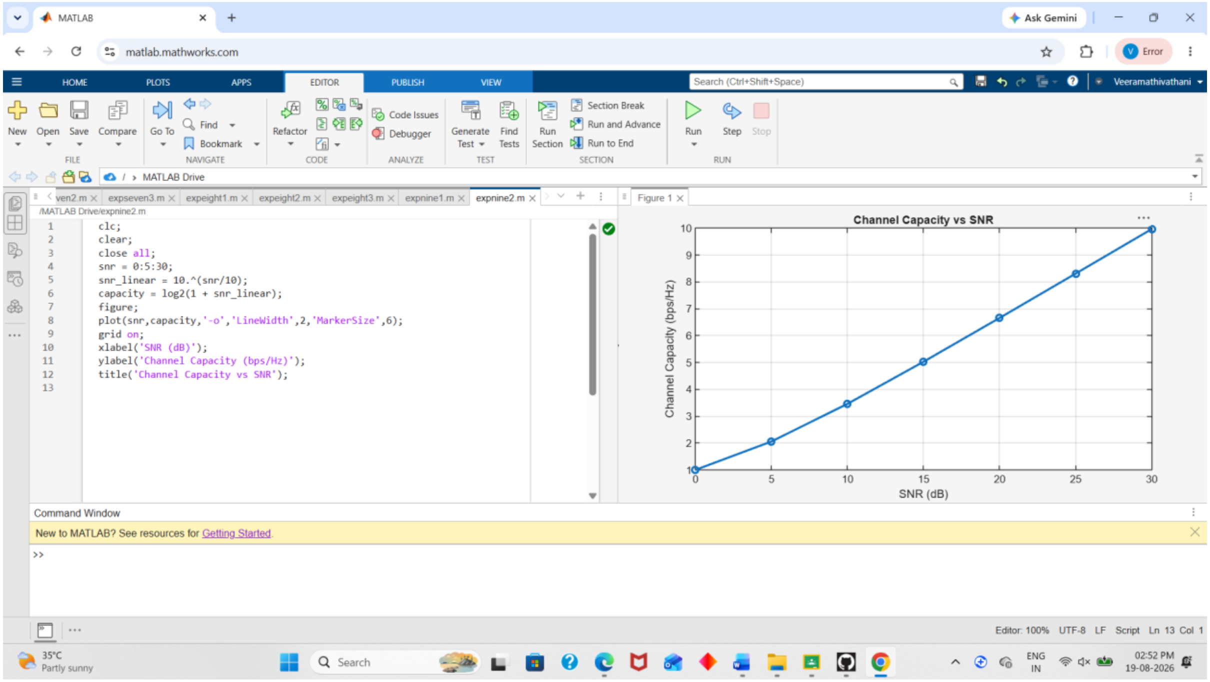
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');



Objective 3: Matlab Code and Output

clc;

clear;

close all;

antennas = [4 8 16 32 64]; % Number of Antenna Elements

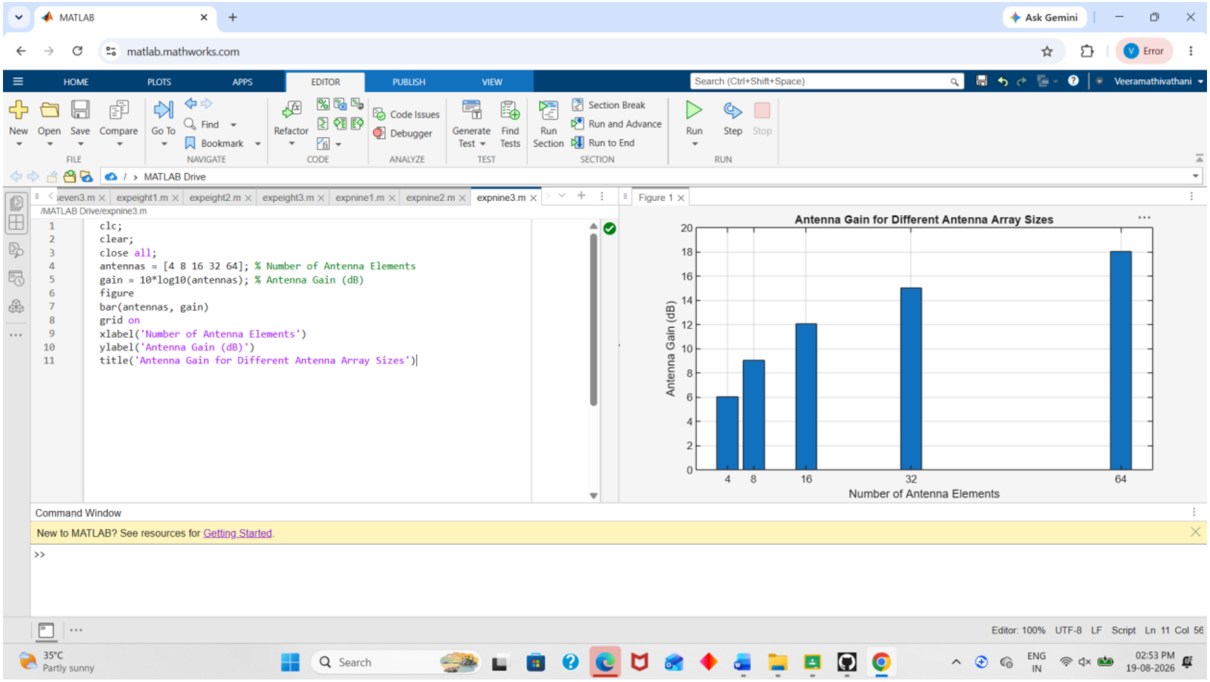
gain = 10*log10(antennas); % Antenna Gain (dB)

figure

bar(antennas, gain)

grid on

```
xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')
```



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.